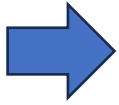


5. Looping (For, While)



Introduction to for and while loops.

➤ it repeats the particular statement till the condition will satisfied.

➤ Types of loop.

1. Entry control

- While loop
- For loop

2. Exit control

➤ Do while loop

1. While loop

➤ A while loop is used to repeat a block of code as long as a given condition is True

➤ Syntax

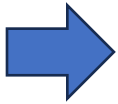
➤ while condition:
 # code to execute repeatedly

2. for loop

- A for loop is used to iterate (repeat) over a sequence like a list, string, tuple, or range.

➤ Syntax

- for variable in range():
 # code to execute



How loops work in Python.

- Loops in Python are used to repeat a block of code multiple times as long as a certain condition is met or over a sequence of items.

1. while Loop

- Repeats a block of code as long as the condition is True.
- You must manually update the variable, or it may run forever.

➤ Syntax

- while condition:
- # code block

2. For loop

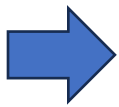
- Iterates over a sequence (like list, tuple, string, or range).

- Automatically stops when the sequence ends.

➤ Syntax

- `for variable in range():`

`# code to execute`



Using loops with collections (lists, tuples, etc.).

- you can use loops to iterate through collections like lists, tuples, sets, strings, and dictionaries.

1.list

- A list is a built-in data type in Python used to store multiple items in a single variable.

➤ `fruits = ["apple", "banana", "cherry"]`

```
for fruit in fruits:  
    print(fruit)
```

2.tuple

- A tuple is a built-in data type in Python used to store multiple items in a single variable, just like a list — but it is immutable

➤ Key Features:

- Ordered → items have a fixed order.
- Immutable → items cannot be changed.
- Allows duplicate values.
- Can store different data types.

➤ `colors = ("red", "green", "blue")`

`for color in colors:`

`print(color)`